

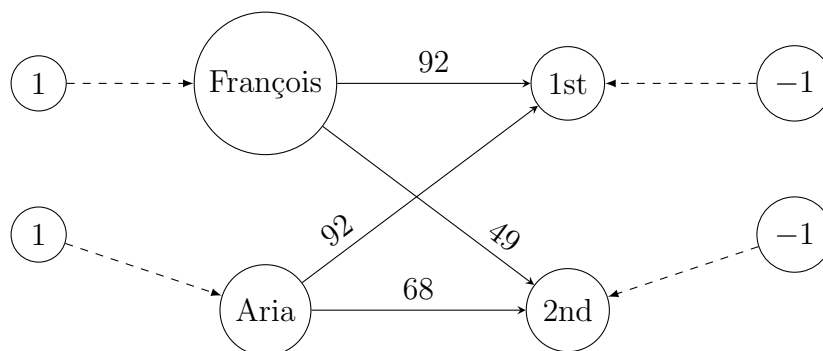
Modeling: assignment problem

Michel Bierlaire

Solution of the practice quiz

The problem is not associated with a physical network. Still, we define a network in the following way. We define a node for each ticket, represented on the right of the network below, and a node for each traveler, represented on the left.

We introduce an arc between each pair of traveler and ticket. For example, the arc from François to “1st” means that François would buy the first class ticket. The cost associated with each arc is the price of the corresponding transaction.



A transshipment problem is written

$$\min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}.$$

subject to the flow conservation constraints,

$$\text{div}(x)_i = s_i \quad \forall i \in \mathcal{N},$$

and the bounds on the flows:

$$\ell_{ij} \leq x_{ij} \leq u_{ij}, \quad \forall (i, j) \in \mathcal{A}.$$

In our case,

- $\mathcal{N}$ is the set of nodes: François (F), Aria (A), 1st (1) and 2nd (2),
- $\mathcal{A}$ is the set of the four possible transactions.
- Costs: c_{ij} , $\forall (i, j) \in \mathcal{A}$ are the transaction costs as reported in the problem description.
- Supply: For each traveler, there is only one transaction. Therefore, the supply is 1. Similarly, each ticket will be purchased by only one travel, so the demand is -1.
- Bounds: As the transactions can only occur in one direction, we impose $x_{ij} \geq 0$, $\forall (i, j) \in \mathcal{A}$. No upper bound is necessary.

The transshipment problem is

$$\min_{x \in \mathbb{R}^6} 92x_{F1} + 49x_{F2} + 92x_{A1} + 68x_{A2},$$

subject to

$$\begin{aligned} x_{F1} + x_{F2} &= 1, \\ x_{A1} + x_{A2} &= 1, \\ -x_{F1} - x_{A1} &= -1, \\ -x_{F2} - x_{A2} &= -1, \\ x_{ij} &\geq 0, \quad \forall (i, j) \in \mathcal{A}. \end{aligned}$$

If we number the variables in the following way:

$$\begin{aligned} x_1 &= x_{F1}, & x_2 &= x_{F2}, \\ x_3 &= x_{A1}, & x_4 &= x_{A2}, \end{aligned}$$

and multiply some constraints by -1 to obtain a right hand side that is non negative, we obtain the following data:

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, b = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

and

$$c^T = (92 \quad 49 \quad 92 \quad 68).$$

An auxiliary variable is associated with each of the equality constraints $(x_5, \dots, x_8)$ and the Phase I can be started.

Iteration 0

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	
1	1	0	0	1	0	0	0	1
0	0	1	1	0	1	0	0	1
1	0	1	0	0	0	1	0	1
0	1	0	1	0	0	0	1	1
-2	-2	-2	-2	0	0	0	0	-4

Iteration 1

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	
1	1	0	0	1	0	0	0	1
0	0	1	1	0	1	0	0	1
0	-1	1	0	-1	0	1	0	0
0	1	0	1	0	0	0	1	1
0	0	-2	-2	2	0	0	0	-2

Iteration 2

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	
1	1	0	0	1	0	0	0	1
0	1	0	1	1	1	-1	0	1
0	-1	1	0	-1	0	1	0	0
0	1	0	1	0	0	0	1	1
0	-2	0	-2	0	0	2	0	-2

Iteration 3

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	
1	1	0	0	1	0	0	0	1
-1	0	0	1	0	1	-1	0	0
1	0	1	0	0	0	1	0	1
-1	0	0	1	-1	0	0	1	0
2	0	0	-2	2	0	2	0	0

Iteration 4

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	
1	1	0	0	1	0	0	0	1
-1	0	0	1	0	1	-1	0	0
1	0	1	0	0	0	1	0	1
0	0	0	0	-1	-1	1	1	0
0	0	0	0	2	2	0	0	0

The optimal value is zero, and we have found a feasible solution of the original problem. The auxiliary x_8 is in the basis, and cannot be pivoted out. It corresponds to the redundant constraint, that can be removed. After cleaning and calculation of the reduced costs, we obtain the following tableau to start Phase II:

x_1	x_2	x_3	x_4	
1	1	0	0	1
-1	0	0	1	0
1	0	1	0	1
19	0	0	0	-141

It happens to be optimal and the optimal solution is

$$\begin{aligned} x_1 = x_{F1} &= 0, & x_2 = x_{F2} &= 1, \\ x_3 = x_{A1} &= 1, & x_4 = x_{A2} &= 0, \end{aligned}$$

for a total cost of 141. This corresponds to the solution where François takes the second class ticket, and Aria the first class. Note that the solution is integer, while this has not been explicitly required.